Lesson 32: The Research Process: Topic and Research Question

English Stage 3 (C1) · Unit 4: Research and Evidence

Adapted from Reading, Writing, Research, and Reasoning: An Advanced ESL Text by Gabriel Winer and Elizabeth Wadell, ASCCC Open Educational Resources Initiative (CC BY-NC 4.0): 3.4 Research Process; 3.5 Finding Your Topic and Research Question.

In this lesson

You learn that research is a messy, repeated process rather than a straight line, and you follow how Lily found, changed and narrowed her topic. You practise choosing and narrowing a topic and writing and evaluating research questions, and you write the research question for your own project. A good research question keeps your reading focused and later becomes your thesis.

Learning goals

After this lesson you should be able to:

- Describe research as an iterative and recursive process.
- Use key questions, sentence frames, freewriting and an internet search to find and narrow a topic.
- Write research questions that begin with Why, How or What are the most significant …
- Evaluate research questions with a four-question checklist.
- Write a focused research question for your own project.

Key language

Word or phrase	Meaning	Example
linear	in a straight line, one step after another	"Steps of the research process in a straight line"
iterative	repeated again and again, so that your thinking and topic keep improving	Research is an iterative process.
recursive	going back and forth between steps	You go back and forth between writing, thinking, reading and researching.
broaden	make a topic wider and more general	She broadened her topic to all kinds of colleges.
narrow (down)	make a topic smaller and more specific	narrow "the effects of teachers on students"

Word or phrase	Meaning	Example
too broad	covering so much that you cannot say anything specific	"The effects of teachers on students" is too broad.
freewriting	writing without stopping for a set time, without worrying about correctness	Set a timer for 5 minutes and freewrite.
refine	improve by making small changes	refine your research question
background	general information that helps readers understand a topic	She could use that information as background.
simple fact	information that answers a question completely, such as a number	How many professors in the US are minorities?
What are the most significant [causes/effects/problems/reasons/solutions] ...?	a frame for a research question	What are the most significant causes of stress for international students?
Although many people think X, I think Y.	a frame for finding a topic	Although many people think all-nighters help, I think sleep matters more.

Rules and explanations

Research is a process, not a straight line

- The usual steps: choose and narrow a topic, write and refine research questions, search for and read articles, then write the paper with those sources.
- In practice, the order changes. If you have no topic yet, you may start by searching and reading. New information may change your opinion or your topic; even professional researchers change their minds midway.
- Research is **iterative**: your thinking keeps changing and your topic keeps improving as you read, write and think. It is also **recursive**: you go back and forth between writing, thinking, reading and researching. The prefix *re-* means "again": **research means to search again**.
- Keep an open mind: explore new ideas first, then form your opinions.

Choosing a topic

- Choose a topic that you personally care about, because you will spend weeks on it. Research is a conversation: choose a topic where you have something new to contribute, not just your first idea or a "good research topic" you have heard about.

- **Key questions** (full list in the lesson material), for example: What have I studied? What makes me mad? What do I think is unfair? If I could change a rule, policy, law, practice or attitude, what would it be?
- **Fill in the blanks:** Although many people think X, I think Y. / I would really like to change X. / More countries should do X like it is done in [another place]. / X is more important than most people realize.

Narrowing a topic

- A topic should be narrow enough to say something unique, but broad enough to find published information. "The effects of teachers on students" is too broad: there are too many kinds of teachers, students and effects.
- Picture four circles inside each other: all possible topics, the assigned topic, the topic narrowed by your first exploration, and (smallest) the research question.
- Three techniques: **freewriting** (5 minutes, in English or another language; then ask what interests you and why); **writing questions** about each part of the topic (What kinds of effects? Which students? Which teachers?); **an internet search** (the top 20 or 30 results give ideas, even if those sources are not good enough for the final paper).

Writing and evaluating a research question

A research question guides your research and keeps you focused. Your thesis statement will usually be its answer.

- You do not already know the answer, and it cannot be answered with a simple fact.
- Useful beginnings: **Why ...? How ...? What are the most significant [causes/effects/problems/reasons/solutions] ...?**
- It can have two parts: first why something is a problem, then how to solve it.

Checklist: (1) Is it narrow and focused? (2) Is it broad enough that you can find enough information? (3) Is it something that you do not already know the answer to? (4) Is it something that cannot be answered with a simple fact?

> **Psychology note:** A psychology research question usually names the people studied (the population) and the things it connects (the variables), for example: How does **the amount of sleep** affect **memory** in **university students**?
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> **Note for Arabic speakers** - Arabic forms questions without a helping verb like *do*, so a common error is "How sleep affects memory?" In a direct question, put *do/does/did* or another auxiliary before the subject: "How **does** sleep affect memory?" "Why **are** there so few professors of color?"

Practice

Exercise 1. Apply this! (3.4) Think about the last writing assignment you were proud of. Draw your writing process. Does it look more like a straight line or a messy circle? What are some advantages of

writing in a messy circle?

Exercise 2. Lily's research path. Read 3.4 and 3.5 in the lesson material.

1. What did Lily read that gave her the idea for her topic?
2. What did she first want to research, and why did she change her topic?
3. Why was it lucky that she had not started writing her essay?
4. Why did she drop her first question, "Are there many faculty of color ... at colleges?" How could she still use its answer?
5. Which two questions did she decide to focus on?
6. Explain in your own words: "Research means to search again."

Exercise 3. Finding your topic. Apply this! (3.5) Choose one technique: the key questions or the fill-in-the-blank sentences. Set a timer for 10 minutes and write as many ideas as you can. Then circle the ideas that interest you most.

Exercise 4. Narrowing a topic. A. Put these in order, from the biggest circle to the smallest. a. How does the amount of sleep affect university students' memory of what they study? b. all possible topics c. psychology (the topic your instructor assigns) d. sleep and memory in university students

B. The book narrows "the effects of teachers on students" by asking about each part. Do the same for "the effects of sleep on students": write one question about the effects, one about the students and one about the sleep.

C. Apply this! (3.5) Search the internet for "effects of teachers on students" and spend 5 minutes on the top 20 or 30 results. (The book prints "effects of students on teachers", but its topic and its list of results are about the effects of teachers on students.) Then search for your own topic and write down five narrower topics that you find.

Exercise 5. Which question is strongest? (3.5) Lily wrote these questions. Which one is the strongest? Why? Use the checklist.

1. How many teachers and professors in the US are minorities?
2. Does Laney College have many faculty of color?
3. How does having a teacher of color affect students?
4. Why is it important to have faculty of color in colleges and what can colleges do to increase the number of faculty of color?

Exercise 6. Strong or weak? Use the checklist. If a question is weak, say which check it fails.

1. How many hours do university students sleep each night?
2. Is sleep important?
3. How does the amount of sleep affect memory and learning in university students, and what can students do to sleep better?
4. What are the effects of sleep?
5. Why do many first-year university students sleep less than they need, and what are the most significant effects on their exam results?

6. Did I sleep enough last night?

Exercise 7. Your research question. Write three research questions for your own topic. Check each one with the checklist and choose the best.

Speak with your tutor

- "I want to choose a research topic. Ask me the key questions from Lesson 32 one at a time, and help me find what I really care about."
- "My topic is [topic]. Ask me questions about each part of it, as the book does with 'the effects of teachers on students', until it is narrow enough."
- "Act as a librarian. I will explain my topic and research question; ask me questions and suggest how to focus it."
- "Here are my three research questions: [paste]. Check each one with the four-question checklist and correct the word order of the questions."

Read aloud

Stressed syllables are in CAPITALS. Wh- questions usually end with falling intonation.

1. Research is an IT·er·a·tive process.
2. Research is also a re·CUR·sive process.
3. Research means to search again.
4. Why is it important to have more minority faculty in colleges?
5. How can we improve this situation?
6. Although many people think that all-nighters help, / I think sleep is more important for memory.

Writing task

Write a research proposal paragraph (150 to 250 words) for your project. Topics: your own; from psychology, sleep and memory in university students, or stress and coping among international students; or the book's topic, teachers who share their students' background.

Include: your topic and why you care about it; how you narrowed it; what you already know; your research question (one or two parts); and what information you need to find (background facts, research studies, examples).

Self-editing checklist:

- Is my topic something I really care about?
- Did I show how I narrowed it from a broad topic?
- Does my research question begin with Why, How or What are the most significant ...?
- Does it pass all four checks: focused, enough information, answer unknown, not a simple fact?
- Is the question correctly formed (for example, "How does ... affect ...?")?
- Did I say what kinds of information I need to find?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- OpenStax *Psychology 2e* (free textbook), chapter 4, States of Consciousness (sleep): https://openstax.org/books/psychology-2e/pages/4-introduction
- OpenStax *Psychology 2e*, chapter 8, Memory: https://openstax.org/books/psychology-2e/pages/8-introduction

Answer key

Exercise 1 (example answer) "A messy circle: while reading, I changed my main point and rewrote my introduction. Advantages: you can use new information, and your topic improves as you learn."

Exercise 2

1. Her professor suggested reading a newspaper such as The New York Times, and she found an article about Nikole Hannah-Jones, a famous journalist denied tenure (a permanent position) at the University of North Carolina; many believed she faced discrimination as a Black woman. It made Lily curious about other professors of color and reminded her that almost all her own teachers were white.
2. The importance of professors of color for community colleges in California. She could not find anything about it, so she broadened the topic to all kinds of colleges in the whole United States.
3. If she had started writing, she would have had to start all over again.
4. A little research answered it: faculty of color are underrepresented. A question with a known, factual answer is not a good research question, but she could use the answer as background for one of her other questions.
5. "Why is it important to have more minority faculty in colleges? How can we improve this situation?"
6. (example answer) Research is not done once: you search, read and think again and again, and your topic and question change as you learn.

Exercise 3 (example answers) "Although many people think that staying up all night helps you prepare for an exam, I think sleep is more important for memory." "Sleep is more important for learning than most students realize."

Exercise 4 A. b, c, d, a. B. (example answers) Effects: am I interested in effects on memory, mood or grades? Students: which students (age, school or university, first-year or international students)? Sleep: the amount of sleep, the quality of sleep, naps, or the time students go to bed? C. (example answers, for "effects of sleep on students") all-nighters before exams; phone use in bed; sleep and exam grades; sleep, mood and stress; naps and learning.

Exercise 5 Question 4 is the strongest: it is focused on colleges, has enough published information and an answer Lily does not know yet, cannot be answered with a simple fact, begins with "Why", and has two parts (why there is a problem, then how to solve it). Question 1 can be answered with a simple fact (a statistic). Question 2 is a yes/no question about one college: a simple fact, with little published information. Question 3 begins with "How" and is not a simple fact, but it is too broad (teachers of color at any level of education, and all kinds of students and effects).

Exercise 6

1. Weak: it can be answered with a simple fact (a number).
2. Weak: a yes/no question whose answer you already know; also too broad.

3. Strong: focused, enough published research, not a simple fact; two parts (a problem and a solution).
4. Weak: too broad (all effects, all people).
5. Strong: focused (first-year students, exam results), not a simple fact; it uses "Why" and "What are the most significant ...".
6. Weak: it is only about you, so there is no published information, and the answer is a simple fact.

Exercise 7 (example answer, the guides' psychology project) "How does the amount of sleep affect memory and learning in university students, and what can students do to sleep better?" It passes all four checks.

My notes